

ARDABIL, Iran, Islamic Republic of

WMO: 407080

Lat: 38.2187N

Lon: 48.3289E

Elev: 1332

StdP: 86.31

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-13.8	-19.8	0.8	-17.8	-15.8	1.1	-11.9	17.9	6.6	15.6	6.3	0.9	300	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.3	30.1	17.2	28.2	16.9	26.8	16.6	20.3	25.9	19.2	24.8	18.2	23.5	7.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	15.7	23.1	17.4	14.6	21.7	16.4	13.7	20.7	65.1	26.0	60.7	25.0	57.5	23.5	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.8	11.8	-23.5	35.8	3.3	2.0	-25.8	37.2	-27.7	38.4	-29.6	39.5	-32.0	41.0		
(5)			-24.3	22.2	4.0	1.9	-27.2	23.6	-29.5	24.7	-31.8	25.8	-34.7	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-1.1	0.5	5.0	8.8	13.8	16.7	18.5	18.6	15.5	11.3	5.3	0.5		
	DBStd	8.19	5.88	5.35	5.38	4.65	3.21	2.49	2.47	2.55	3.04	3.95	4.45	5.63		
	HDD10.0	1343	346	266	172	76	6	0	0	0	1	32	149	296		
	HDD18.3	3324	604	499	414	288	144	61	28	28	94	221	391	553		
	CDD10.0	1154	0	1	16	38	124	202	262	266	166	71	8	1		
	CDD18.3	94	0	0	1	1	4	12	32	35	8	1	0	0		
	CDH23.3	1536	0	0	4	31	119	248	379	519	182	53	0	0		
	CDH26.7	388	0	0	0	5	22	56	97	146	56	6	0	0		
(14) Wind	WSAvg	5.0	5.1	4.8	5.4	5.0	4.9	5.4	5.8	5.6	5.0	4.5	4.1	4.4		
(15) Precipitation	PrecAvg	325	26	26	35	39	47	18	9	9	18	36	38	24		
	PrecMax	458	47	58	83	93	88	61	33	27	60	113	126	65		
	PrecMin	235	6	5	0	1	19	0	0	0	0	0	0	0		
	PrecStd	56	11	14	18	24	19	18	8	8	16	27	30	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	15.2	23.0	26.9	29.1	31.2	32.4	32.9	31.5	27.7	19.9	15.9		
		MCWB	5.9	7.2	10.1	13.5	15.4	16.4	18.5	18.1	15.5	13.1	9.9	7.9		
	2%	DB	10.8	13.0	19.9	23.0	26.2	28.1	29.1	30.0	28.1	24.9	17.4	13.1		
		MCWB	4.6	5.8	9.4	12.2	14.7	16.5	18.1	18.0	15.9	12.7	8.9	6.3		
	5%	DB	9.0	10.9	17.1	20.1	24.1	26.1	27.0	28.0	25.2	22.0	15.1	10.9		
		MCWB	3.8	5.0	8.1	11.1	14.4	16.4	17.8	17.7	15.9	12.5	8.2	5.2		
	10%	DB	7.1	8.8	14.5	17.9	22.0	24.2	25.1	26.2	22.3	19.1	13.0	8.9		
		MCWB	2.7	4.1	7.2	10.5	13.5	15.8	17.4	17.0	15.6	11.8	7.6	4.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	8.0	11.2	15.2	18.4	19.9	22.0	22.5	19.6	16.3	11.8	8.7		
		MCDB	11.7	14.0	19.7	23.1	24.0	26.2	28.1	28.4	25.0	21.4	16.8	13.9		
	2%	WB	5.2	6.5	9.8	13.3	16.3	18.5	20.3	20.4	18.1	14.6	10.2	6.7		
		MCDB	9.5	11.6	18.9	21.0	22.6	24.9	26.0	25.9	22.9	19.2	14.8	11.9		
	5%	WB	4.0	5.4	8.6	12.0	15.1	17.5	19.1	19.3	17.2	13.6	9.1	5.5		
		MCDB	8.1	9.7	16.3	18.8	21.8	23.6	24.8	24.7	21.7	18.6	13.5	9.9		
	10%	WB	2.9	4.2	7.5	10.9	14.2	16.5	18.0	18.3	16.3	12.6	8.0	4.3		
		MCDB	6.7	8.3	13.9	16.7	20.5	22.4	23.2	23.4	20.3	17.8	12.2	8.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	11.4	12.5	13.3	13.6	13.3	12.1	13.3	12.1	12.1	11.8	11.6		
		MCDBR	12.0	14.1	17.1	18.4	18.1	17.7	16.3	18.2	19.5	18.5	16.7	14.4		
		MCWBR	7.4	8.4	9.0	9.4	8.8	9.1	8.7	9.8	10.1	9.2	9.7	8.9		
		MCDBR	11.4	12.8	15.9	16.4	16.4	15.0	13.9	14.5	14.6	14.6	13.3	12.7		
	5% WB	MCWBR	7.3	8.1	8.9	9.0	9.2	8.9	8.3	8.8	8.6	8.4	8.3	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.339	0.389	0.458	0.452	0.444	0.479	0.451	0.413	0.389	0.332	0.303		
	taud		2.353	2.261	2.141	1.986	2.055	2.112	2.030	2.110	2.197	2.218	2.350	2.383		
	Ebn at Noon		870	879	864	823	834	840	806	820	830	809	828	845		
	Edh at Noon		90	113	141	175	166	157	169	153	132	117	90	82		
(44) All-Sky Solar Radiation	RadAvg		2.16	2.94	3.73	4.68	6.00	7.17	6.99	6.32	4.87	3.35	2.32	1.90		
	RadStd		0.20	0.41	0.42	0.53	0.35	0.40	0.36	0.44	0.34	0.36	0.29	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.33	N/A	N/A	+0.97	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page